Artificial Intelligence for Interpreting Occupational Health Examination Tests: A Systematized Review of Evidence and Gaps

Supplement to the e-poster, 1st PERDOKI Jakarta Summit 2026. Vidya Cecilia, Occupational Medicine Residency Program, Faculty of Medicine, Universitas Indonesia.

Included studies (n = 13)

1. Heo SJ, Kim Y, Yun S, Lim SS, Kim J, Nam CM, et al. Deep learning algorithms with demographic information help to detect tuberculosis in chest radiographs in annual workers' health examination data. Int J Environ Res Public Health. 2019;16(2):250. doi: 10.3390/ijerph16020250
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3. Young C, Barker S, Ehrlich R, Kistnasamy B, Yassi A. Computer-aided detection for tuberculosis and silicosis in chest radiographs of gold miners of South Africa. Int J Tuberc Lung Dis. 2020;24(4):444-51. doi: 10.5588/ijtld.19.0624
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6. Wang Y, Chen Z, Tian S, Zhou S, Wang X, Xue L, et al. Convolutional neural network-based ECG-assisted diagnosis for coal workers. Int J Environ Res Public Health. 2022;20(1):9. doi: 10.3390/ijerph20010009
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9. He L, Luo Y, Lei J, Yin Q, Yang D, Zhang C, et al. [Evaluation of clinical value of interpretable Transformer-based models in computer-aided diagnosis of early-stage pneumoconiosis]. J Environ Occup Med. 2026;43(7):844-51. Chinese. doi: 10.11836/JEOM25477
10. Vella F, Senia P, Filetti V, Vitale E, Sferrazzo G, Privitera L, et al. Application of large language models as decision support tools in occupational health and safety management: a cohort study of industrial workers. Front Public Health. 2026;14:1815316. doi: 10.3389/fpubh.2026.1815316
11. Moore BCJ, Schlittenlacher J. Diagnosing noise-induced hearing loss sustained during military service using deep neural networks. Trends Hear. 2023;27:23312165231184982. doi: 10.1177/23312165231184982
12. Priego-Torres B, Sanchez-Morillo D, Khalili E, Conde-Sánchez MÁ, García-Gámez A, León-Jiménez A. Automated engineered-stone silicosis screening and staging using deep learning with X-rays. Comput Biol Med. 2025;191:110153. doi: 10.1016/j.compbiomed.2025.110153
13. Dong H, Zhu B, Zhang X, Kong X. Use data augmentation for a deep learning classification model with chest X-ray clinical imaging featuring coal workers' pneumoconiosis. BMC Pulm Med. 2022;22(1):271. doi: 10.1186/s12890-022-02068-x

Study characteristics and main results

#	Study	Country	Modality / target	Reference standard	Main results
1	Heo 2019 (PMID 30654560)	South Korea	CXR / tuberculosis	Radiologist reading under the national QA programme	Test AUC VGG19 0.9075 (image-only) to 0.9213 (+ demographics), p < 0.05
2	Wang 2020 (PMID 32471837)	China	CXR / pneumoconiosis vs normal	Expert panel (1 certified radiologist + 4 occupational physicians), ILO	AUC 0.878 (95% CI 0.811-0.946) vs radiologists 0.668 and 0.772
3	Young 2020 (PMID 32317070)	South Africa	CXR / silicosis, TB, silicotuberculosis	Radiologist + 3 physicians (board); 4 expert readers (screening set)	Best system AUC 0.986 for any abnormality (Se 98.2, Sp 98.2); TB vs normal 0.989; silicosis vs normal 0.986 (system A), 0.939 (B)
4	Charapaqui-Miranda 2020 (Scopus 85084850887)	Peru	Multimodal exam data / work-entry MCU fit/unfit category (fit or fit with modifications vs unfit)	Fit/unfit result recorded by the examining doctor (MCU status, not a 7-step fitness-to-work assessment)	Best model logistic regression: AUC 60.4%, accuracy 68.1%
5	Ehrlich 2022 (PMID 36231700)	South Africa	CXR / silicosis, TB, any abnormality	2 occupational medicine specialists (reader 2 = consensus of two)	Silicosis AUC 0.883 and 0.839 (ILO >= 1/0), 0.887 and 0.892 (>= 1/1) vs reader 1 and 2 (system A)
6	Wang 2022/2023 (PMID 36612331)	China	12-lead ECG / 4 abnormal ECG classes	ECG diagnosis from the examination (cardiologist-performed ECG; labelling procedure not stated explicitly)	Test accuracy 93.0-97.7%, AUC 0.92-0.98 across 4 ECG classes
7	Li 2024 (PMID 38348336)	China	CXR / pneumoconiosis vs normal	10 occupational disease experts + 2 radiologists, 3-5 labellers per film	Validation set (261 pneumoconiosis, 234 non): accuracy 95%, AUC 0.947
8	Liu 2026 (PMID 42457633)	USA	CXR / 4 ILO-aligned tasks (1-2 screening, 3-4 grading)	Consensus of certified B-readers (2 + third on disagreement)	In-distribution accuracy 0.93, 0.92, 0.91 (tasks 1-3), 0.83 (task 4) vs B-readers 0.90, 0.91, 0.77, 0.86
9	He 2026 (Scopus 105046943496)	China	CXR / early pneumoconiosis, binary	Consensus of 3 senior radiologists	CrossFormer AUC 0.962 (95% CI 0.945-0.979), accuracy 92.0%, F1 0.916
10	Vella 2026 (PMID 42180469)	Italy	Occupational health records / risk assessment, surveillance protocol, fitness judgements from documents only (4 categories; not a 7-step fitness-to-work assessment)	Experienced occupational physician	Perfect concordance of document-based fitness judgments between LLM and physician
11	Moore 2023 (PMID 37550005)	UK	Pure-tone audiometry 1-8 kHz both ears + age / military noise-induced hearing loss	Group membership: claimant with medical report excluding other causes vs unexposed control (no individual diagnosis)	MLP(18) on test databases: Se 98.6, Sp 90.2, d' 3.49 (rule-based methods Sp 40-64)
12	Priego-Torres 2025 (PMID 40252290)	Spain	CXR / silicosis screening (healthy vs silicosis) and staging (SS vs PMF)	Clinical diagnosis: exposure history, ILO reading and HRCT (ICOERD), some biopsy; controls labelled normal in VinDr	Screening AUC 1.00, accuracy 0.97-0.99 (4 CNNs); staging SS vs PMF accuracy 0.81-0.84, AUC 0.91-0.94
13	Dong 2022 (PMID 35840945)	China	CXR / CWP radiographic features (nodules, interstitial change, emphysema)	Reports from 3 radiologists, reviewed by 2 senior radiologists	Average AUC 0.98 (nodules 0.97, interstitial 0.96, emphysema 1.00); accuracy 0.96

JBI diagnostic test accuracy checklist (item level; no overall score)

#	Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Yes
1	Heo 2019	Y	Y	U	Y	N	U	Y	Y	Y	U	6/10
2	Wang 2020	U	N	U	Y	U	Y	Y	Y	Y	U	5/10
3	Young 2020	N	N	N	Y	N	Y	Y	Y	Y	U	5/10
4	Charapaqui-Miranda 2020	U	Y	N	Y	NA	U	Y	Y	Y	N	5/10
5	Ehrlich 2022	Y	Y	Y	Y	Y	U	Y	Y	Y	Y	9/10
6	Wang 2023 (ECG)	U	Y	U	Y	U	U	U	Y	U	U	3/10
7	Li 2024	N	N	U	Y	U	U	U	Y	Y	Y	4/10
8	Liu 2026	N	U	U	Y	U	Y	Y	Y	Y	Y	6/10
9	He 2026	N	N	N	Y	U	Y	Y	Y	Y	Y	6/10
10	Vella 2026 (LLM)	U	Y	U	U	NA	U	Y	Y	Y	Y	5/10
11	Moore 2023 (audiometry)	N	N	U	Y	U	N	Y	Y	N	Y	4/10
12	Priego-Torres 2025	N	N	U	Y	NA	Y	Y	U	N	Y	4/10
13	Dong 2022	N	N	N	N	NA	Y	Y	Y	Y	U	4/10

Y yes · N no · U unclear · NA not applicable. Q1 consecutive or random sample; Q2 case-control design avoided; Q3 inappropriate exclusions avoided; Q4 index test read without knowledge of the reference; Q5 threshold pre-specified; Q6 reference standard likely correct; Q7 reference read without knowledge of the index test; Q8 appropriate interval; Q9 same reference for all; Q10 all participants analysed.